

INDIAN STATISTICAL INSTITUTE

End-Semester Examination (2025–26)

Programme: M.Math. 2nd Year

Course: Algebraic Geometry

Date: May 8, 2026

Time: 3 hours

Maximum Marks: 50

Instructions.

- This is **not** an open-notes examination; calculators are not allowed.
- Justify all steps in your arguments. Do not assume results not covered in lectures without providing clear proofs.
- Clarity and logical presentation are essential. Vague answers may lose marks. Partial solutions may receive partial credit.

Notations & Conventions.

- Throughout, k denotes an algebraically closed field unless otherwise specified.

(1) Prove the following:

- (a) $\mathbb{A}_k^1$ is not isomorphic to any proper open subset of itself. [5 marks]
- (b) $\mathbb{A}_k^2$ is not homeomorphic to $\mathbb{P}_k^2$. [3 marks]
- (c) If an affine variety is isomorphic to a projective variety over a field k , then it consists of only one point. [2 marks]

(2) (a) Let $F \in k[X_0, \dots, X_n]$ be a homogeneous polynomial of degree d , and let $V = \mathcal{Z}(F) \subset \mathbb{P}_k^n$ be the corresponding projective hypersurface. Assume that $\text{char}(k) \nmid d$. Show that

$$\text{Sing}(V) = \mathcal{Z}\left(\frac{\partial F}{\partial X_0}, \dots, \frac{\partial F}{\partial X_n}\right).$$

Give an explicit example to show that the above equality may fail if $\text{char}(k)$ divides d .

[3 + 2 = 5 marks]

(b) For every degree $d > 0$, and for every $p = 0$ or a prime number, give the equation of a non-singular projective curve of degree d in $\mathbb{P}_k^2$ over a field k of characteristic p . [5 marks]

(3) (a) Let $X \subset \mathbb{A}_k^n$ be the hypersurface defined by a polynomial $f(x_1, \dots, x_n)$. Show that $\mathbb{A}_k^n \setminus X$ is an affine variety. Determine its coordinate ring up to isomorphism of k -algebras. [5 marks]

(b) Prove that $GL_n(k)$ is a linear algebraic group over k . Determine its Zariski tangent space at an arbitrary point $g \in GL_n(k)$. [5 marks]

(4) Assume that $\text{char}(k) = 0$.

(a) Let $C \subset \mathbb{A}_k^2$ be the nodal curve defined by $x^6 + y^6 = xy$. Let

$$\pi : \widetilde{\mathbb{A}_k^2} = \text{Bl}_O(\mathbb{A}_k^2) \rightarrow \mathbb{A}_k^2$$

be the blow-up of $\mathbb{A}_k^2$ at the origin $O = (0, 0)$, and let $\widetilde{C}$ denote the proper transform of C in $\widetilde{\mathbb{A}_k^2}$. Show that $\widetilde{C}$ is non-singular. [6 marks]

(b) Is it true that if $C \subset \mathbb{A}_k^2$ is an affine plane curve whose only singularity is at the origin $O = (0, 0)$ and has multiplicity 2 at O , then the proper transform of C in $\text{Bl}_O(\mathbb{A}_k^2)$ is non-singular? Justify your answer. [4 marks]

(5) Prove that $\mathbb{P}_k^1 \times \mathbb{P}_k^1$ is birationally equivalent to $\mathbb{P}_k^2$, but these two surfaces are not isomorphic as k -varieties. [4 + 6 = 10 marks]

- (6) (a) Let $S \subset \mathbb{P}_k^3$ be an irreducible surface of degree d , and let $C \subset \mathbb{P}_k^3$ be an irreducible curve. Compute the divisor class group

$$\text{Cl}(\mathbb{P}_k^3 \setminus (S \cup C)).$$

[4 marks]

- (b) Let $S_1, S_2, \dots, S_n \subset \mathbb{P}_k^3$ be irreducible projective surfaces of degrees $d_1, d_2, \dots, d_n$ respectively. Compute the divisor class group

$$\text{Cl}(\mathbb{P}_k^3 \setminus (S_1 \cup S_2 \cup \dots \cup S_n)).$$

[6 marks]

- (7) (a) Let C be an algebraic curve, $P \in C$, and A be a finitely generated k -algebra. Consider the skyscraper sheaf $i_P(A)$ at P with value A . Prove that

$$H^i(C, i_P(A)) \cong \begin{cases} A, & \text{if } i = 0, \\ 0, & \text{if } i > 0. \end{cases}$$

[4 marks]

- (b) Let $X = \mathbb{A}_k^2 \setminus \{(0, 0)\}$. Show that

$$H^i(X, \mathcal{O}_X) \cong \begin{cases} k[x, y], & \text{if } i = 0, \\ k\text{-span} \langle x^i y^j : i, j < 0 \rangle, & \text{if } i = 1, \\ 0, & \text{if } i \geq 2. \end{cases}$$

Hence, deduce that X is not an affine variety.

[6 marks]

— End of Question Paper —

All the best!